
Course Instructor: **Nishat Tasnim**
Course: **CSE103 (Structured Programming)**

Lab: **06**

Topics Today

1. Loop, Nested Loop

Problem 1

Write a program to print natural numbers between 1 and 100.

Problem 2

Write a program to print odd numbers between 1 and 50 in reverse order.

Problem 3

Write a C program to find HCF (Highest Common Factor) of two numbers.

Sample Input	Sample Output
Input 1st number for HCF: 24 Input 2nd number for HCF: 28	HCF of 24 and 28 is : 4

Problem 4

Write a program in C to find the number and sum of all integer between 100 and 200 which are divisible by 9.

Expected Output: Numbers between 100 and 200, divisible by 9 :
108 117 126 135 144 153 162 171 180 189 198
The sum : 1683

Problem 5

Write a program in C to print the Floyd's Triangle according to user input.
User input: 5

```
1
0 1
1 0 1
0 1 0 1
1 0 1 0 1
```

Problem 6

Generate the following pattern using nested for loops according to user input.
User input: 5

```
1
2 2
3 3 3
4 4 4 4
5 5 5 5 5
```

Problem 7

Generate the following pattern using nested for loops.

```
1
2 3
4 5 6
7 8 9 10
11 12 13 14 15
```

Problem 8

Generate the following pattern using nested while loops.

```
1
2 3
4 5 6
7 8 9 10
11 12 13 14 15
```

Problem 9

Write a program in C to find the sum of the series $[x - x^3 + x^5 + \dots]$.

Test Data :

Input the value of x :2

Input number of terms : 5

Expected Output :

The values of the series:

2

-8

32

-128

512

The sum = 410

Problem 10

Write a program in C to find LCM (Least Common Multiple) of any two numbers.

Sample Input	Sample Output
Input 1st number for LCM: 15 Input 2nd number for LCM: 20	The LCM of 15 and 20 is : 60